AP CALCULUS AB

Unit 1 • Lesson 1-5 • Notes (filled in)

# Lesson 1-5 • IVT and EVT

Notes (filled in)

**Learning goals**

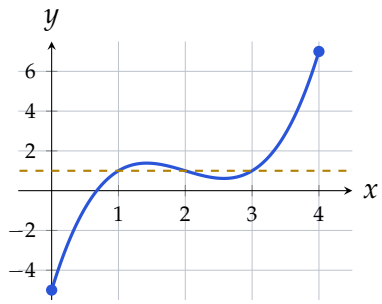
- State the Intermediate Value Theorem and the Extreme Value Theorem, with their hypotheses.
- Use the IVT to justify that an equation has a solution.
- Use the EVT to justify that absolute extrema exist, and find them from graphs and tables.

**Vocabulary.** Intermediate Value Theorem (IVT), Extreme Value Theorem (EVT), closed interval, absolute maximum, absolute minimum, hypothesis, conclusion.

## 1. Intermediate Value Theorem

**IVT**

If $f$ is __continuous__ on the __closed__ interval $[a,b]$ and $k$ is any number __between $f(a)$ and $f(b)$__, then there is at least one $c$ in $[a,b]$ with __$f(c)=k$__.




**Example 1**

Show that $x^3+2x=5$ has a solution in $[1,2]$.

$f(x)=x^3+2x$ is a polynomial, so it is continuous on $[1,2]$. $f(1)=3$, $f(2)=12$. Since $3<5<12$, by the IVT there is a $c$ in $[1,2]$ with $f(c)=5$.

**Example 2 • Table**

$W$ is continuous with $W(0)=8$, $W(2)=3$, $W(5)=6$, $W(9)=1$. What is the least number of solutions of $W(t)=4$ on $[0,9]$?

4 is between 8 and 3, between 3 and 6, and between 6 and 1: at least three.

## 2. Extreme Value Theorem

**EVT**

If $f$ is __continuous__ on the closed interval $[a,b]$, then $f$ has both an __absolute maximum__ and an __absolute minimum__ value on $[a,b]$. They occur at __endpoints__ or at points where the

graph turns (critical points).

**Example 3**

Find the absolute maximum and minimum of $f(x) = x^2 - 4x + 5$ on $[0, 3]$. Why must they exist?

$f$ is a polynomial, continuous on $[0, 3]$, so by the EVT both exist. Candidates: vertex $x = 2$, $f(2) = 1$; endpoints $f(0) = 5$, $f(3) = 2$. Absolute maximum 5 at $x = 0$; absolute minimum 1 at $x = 2$.

**Example 4 • When the theorems do not apply**

Explain why the conclusion can fail: (a) $g(x) = \frac{1}{x}$ on $[-1, 1]$, IVT with $k = 0$; (b) $h(x) = \frac{1}{x}$ on $(0, 1]$, EVT.

(a) $g(-1) = -1$, $g(1) = 1$, yet $g$ is never 0: $g$ is not continuous on $[-1, 1]$.
(b) $h$ has no absolute maximum (it grows without bound near 0): the interval is not closed.

**AP exam tip**

An IVT or EVT justification names the hypothesis ("$f$ is continuous on $[a, b]$ because…"), the values used, and the conclusion.

**Check your understanding**

1. $f$ continuous, $f(1) = 4$, $f(3) = 4$. Does the IVT guarantee $f(c) = 0$ on $[1, 3]$? No.
2. Does a continuous function on $(0, 5)$ always have an absolute maximum? No; the interval must be closed.